Fullstack homework test

🌾 Fullstack Engineer — Homework Test

Context

AgriWatch is a digital platform used by cooperative farms across West Europe to monitor weather conditions and plan agricultural activities. Field agents rely on the platform daily to decide when to sow, irrigate, or harvest.

The product team has received a brief from the Head of Operations:

> "Our field agents waste too much time checking multiple sources for weather. We need a single dashboard where they can monitor conditions for all the plots they manage, get a forecast for the coming days, and set their own preferences. Some agents cover 10+ sites — they need a way to keep track of their key locations quickly."

Your job is to build the **first working version** of this internal tool.

What We Expect

Build a fullstack web application that helps field agents monitor weather across multiple locations.

User Requirements

As a field agent, I need to:

1. **Search for a location** and see its current weather conditions.
2. **View an upcoming forecast** for any given location.

3. **Save locations** I monitor regularly so I can access them without searching every time.
4. **Remove a saved location** when I no longer need to track it.
5. **Set my preferred temperature unit** (Celsius or Fahrenheit) and have it remembered.
6. **Set a default location** that loads automatically when I open the dashboard.
7. **Access the tool from my phone** in the field just as well as from a desktop at the office.
8. **Have my preferences and saved locations tied to my identity** so they persist across sessions and devices.

Stack Guidelines

You can use any tool you need to reach the goal.

AI tooling is welcomed.

Deliverables

1. **A GitHub repository** (public or private with access granted) containing the full source code.
2. **A README** that includes:
   - How to set up and run the project locally.
   - Architecture decisions and trade-offs you made.
   - What you would improve or add given more time.

Evaluation Criteria

Criteria	What we're looking for
Product thinking	Did you make sensible decisions where the spec was vague?
Functionality	Does the core flow work end to end?
Code quality	Clean, modular, consistent — code we'd want to maintain.
Testing	Meaningful coverage, not checkbox coverage.
Documentation	Could a teammate pick this up and run it?
Technical choices	Are they justified? Do they fit the problem?

Timeline

You have **24 hours** from receiving this brief. We don't expect a production-grade product — we expect a thoughtful, working prototype that shows how you think, build, and communicate.

Good luck, and feel free to reach out if something is genuinely blocking — knowing when to ask is a skill too. 🌱